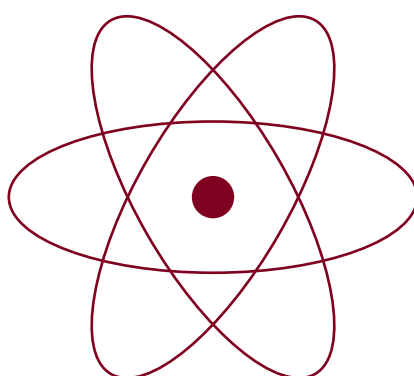


# SPhOH

SAUDI PHYSICS OLYMPIAD HUB

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# Kinematics

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Olympiad Problem Set

Olympiad Training • Mechanics

SPhOH Team

FIRST EDITION • 2026

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# About This Problem Set

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This problem set is part of the training resources of the **Saudi Physics Olympiad Hub (SPhOH)**.

It is designed primarily for students preparing for the Saudi Physics Olympiad and related advanced physics competitions. The purpose of the handout is to provide a focused and substantial collection of problems for deliberate problem-solving practice.

The handout is **problem-centered**. Explanations are intentionally brief and are included only where they help establish the ideas or techniques needed for the problems.

## How to Use the Problem Set

Students are encouraged to attempt each problem independently before looking at its hint. A hint should be treated as a way to overcome a specific obstacle, not as a substitute for solving the problem.

For difficult problems, several progressive hints may be provided. Try to use the least amount of assistance necessary before returning to the problem.

## Difficulty Levels

The normal problems are divided into three training levels:

- **Level I** — Easy and direct substitution problems.
- **Level II** — Problems requiring combinations of ideas and more substantial reasoning.
- **Level III** — National Olympiad level problems.

Find more at: <https://devfhd.github.io/SPhOH/>.

# Kinematics

## Topic Note: Kinematic Relations

Kinematics describes motion without considering the forces causing it. Key relationships include:

$$v_{\text{avg}} = \frac{\Delta x}{\Delta t}, \quad a = \frac{\Delta v}{\Delta t}$$

For uniform acceleration  $a$ , where  $v_0$  and  $v$  represent initial and final velocities while  $x$  and  $\Delta x$  represent position and change in position:

$$v = v_0 + at, \quad x = x_0 + v_0 t + \frac{1}{2}at^2, \quad v^2 = v_0^2 + 2a\Delta x$$

In 2D projectile motion, horizontal and vertical motions are independent:

$$x(t) = (v_0 \cos \theta)t, \quad y(t) = (v_0 \sin \theta)t - \frac{1}{2}gt^2$$

The horizontal range  $R$  and maximum height  $H$  on level ground are given by:

$$R = \frac{v_0^2 \sin(2\theta)}{g}, \quad H = \frac{v_0^2 \sin^2 \theta}{2g}$$

## 1.1 1D Kinematics and Kinematic Equations

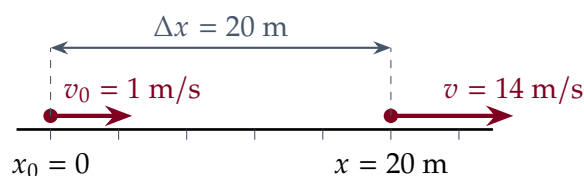
### PROBLEM 1

LEVEL I

#### Acceleration along a Straight Line

Mohammed moves along a straight line with an initial velocity of 1 m/s. His velocity increases uniformly to 14 m/s while covering a distance of 20 m.

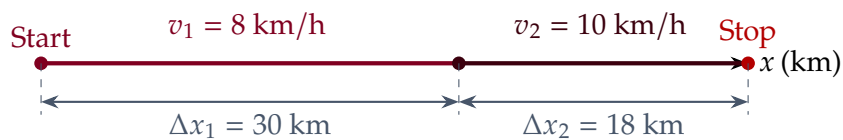
Determine the magnitude and direction of his acceleration.



**PROBLEM 2****LEVEL II****Multi-Stage Average Velocity**

Hadi begins moving along a straight line at a constant speed of  $8 \text{ km/h}$  until he covers a distance of  $30 \text{ km}$ . Immediately following this, he continues moving along the same line at a constant speed of  $10 \text{ km/h}$  for a distance of  $18 \text{ km}$ , after which he comes to a complete stop.

Calculate the average speed for the entire trip (from the start of his motion until he stops).

**PROBLEM 3****LEVEL II****Average Velocity over Split Distance and Time**

A particle covers the first half of its total displacement with a constant velocity  $v_0$ . The second half of the distance is covered in two equal time intervals: moving with a constant velocity  $v_1$  for the first half of that time, and with velocity  $v_2$  for the remaining half of the time.

Calculate the average velocity  $v_{\text{avg}}$  of the particle averaged over the entire duration of its motion.

**PROBLEM 4****LEVEL II****Motorboat and Floating Raft**

A motorboat traveling downstream along a river passes a floating raft at point A. The boat continues downstream for a time interval  $\tau = 60 \text{ min}$ , then turns around and travels upstream at the same power setting (constant speed relative to the water). The boat meets the raft again at a distance  $l = 6.0 \text{ km}$  downstream from point A.

Assuming the river flow velocity remains uniform and constant, determine the speed of the river current.

**PROBLEM 5**

## LEVEL II

**2D Motion under Constant Acceleration**

A particle begins moving from an initial position vector  $\vec{r}_0 = (3\hat{i} - 7\hat{j})$  m with an initial velocity  $\vec{v}_0 = 2\hat{i}$  m/s. It moves with a constant acceleration of  $\vec{a} = (\hat{i} + 3\hat{j})$  m/s<sup>2</sup>.

- (a) Will the particle cross the  $x$ -axis? If so, at what time  $t$  and at what  $x$ -coordinate?
- (b) Will the particle cross the  $y$ -axis? If so, at what time  $t$  and at what  $y$ -coordinate?

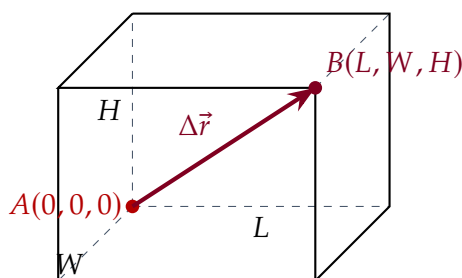
**1.2 Vector Displacement in 3D Space****PROBLEM 6**

## LEVEL II

**3D Box Corner-to-Corner Fly Displacement**

A fly flies inside a rectangular room of dimensions  $L \times W \times H$  from corner  $A$  (at the origin  $(0, 0, 0)$ ) to the diagonally opposite corner  $B$  (at  $(L, W, H)$ ).

Find the magnitude of the fly's displacement vector  $\Delta\vec{r}$ .

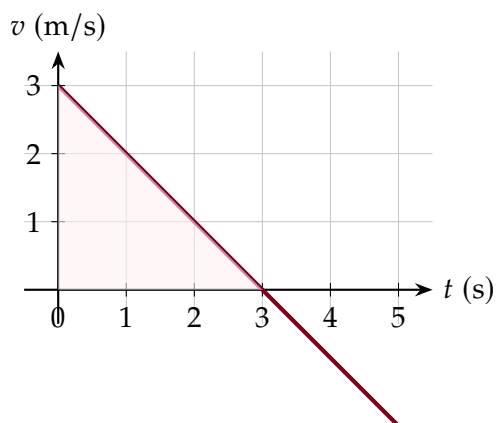
**1.3 Graphical Analysis of Motion****PROBLEM 7**

## LEVEL I

**Position from Velocity–Time Graph**

The graph below represents the velocity  $v(t)$  of an object moving along a straight line as a function of time  $t$ . It is known that the position of the object at the instant when  $v = 0$  is  $x = 4$  m.

Determine the initial position of the object.

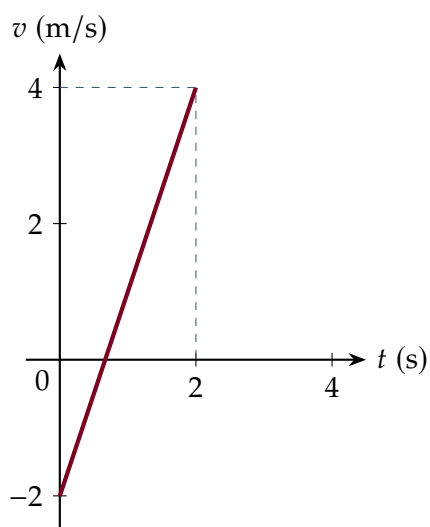


## PROBLEM 8

LEVEL I

### Acceleration from Velocity–Time Graph

The graph below depicts the velocity  $v(t)$  of a particle moving along the  $x$ -axis over time. Determine the instantaneous acceleration  $a$  of the particle at time  $t = 0$ .



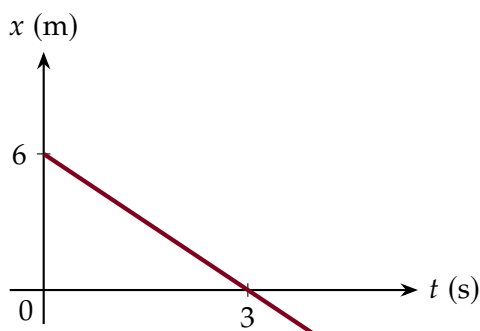
## PROBLEM 9

## LEVEL I

**Instantaneous Velocity from Position–Time Graph**

The position  $x(t)$  of a particle moving along a straight line is plotted as a function of time  $t$  in the figure below.

Determine the instantaneous velocity  $v$  of the object at  $t = 0$ .

**1.4 Free Fall and Vertical Motion under Gravity****PROBLEM 10**

## LEVEL III

**Continuous Drop of Falling Balls**

A child standing on the roof of a residential building releases (drops from rest) tennis balls vertically downward at regular time intervals. The rate of dropping is adjusted such that at the exact instant a new ball is released, the previous ball is at some intermediate position in the air, and the ball before that has just struck the ground.

At the instant a ball is released, what is the height above the ground of the ball currently in the air? The height of the building is  $H = 12$  m. (Take  $g = 9.8 \text{ m/s}^2$  or leave in terms of  $H$ )

**PROBLEM 11**

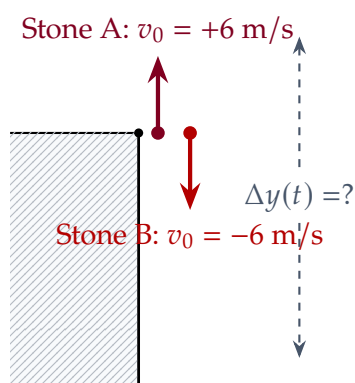
## LEVEL II

**Separation of Oppositely Thrown Stones**

A man standing at the edge of a tall vertical cliff throws two identical stones with the same initial speed  $v_0 = 6 \text{ m/s}$ : Stone A is thrown vertically upward, and Stone B is thrown vertically

downward.

Assuming neither stone has hit the ground below, what is the distance between the two stones after a time  $t = 3$  s? (Take  $g = 9.8 \text{ m/s}^2$ )



## PROBLEM 12

LEVEL II

### Collision of Two Counter-Moving Balls

Two balls are thrown simultaneously along the same vertical line. Ball 1 is launched from the ground vertically upward with an initial speed  $v_1 = 20 \text{ m/s}$ . Ball 2 is released simultaneously from a height  $H = 150 \text{ m}$  above the ground, thrown vertically downward with an initial speed  $v_2 = 30 \text{ m/s}$ . (Take  $g = 10 \text{ m/s}^2$ )

- Calculate the time  $t_{\text{coll}}$  elapsed until the two balls collide, and determine the height  $y_{\text{coll}}$  above the ground at which the collision occurs.
- Which event occurs first: Ball 1 reaching its maximum height, or the collision between the two balls?

## 1.5 Projectile Motion

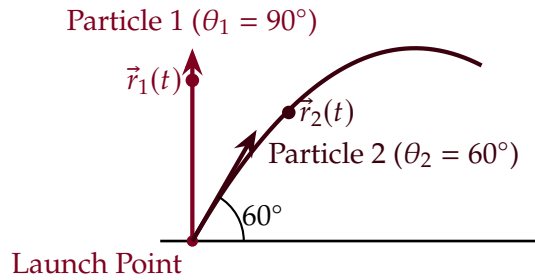
## PROBLEM 13

LEVEL II

### Diverging Projectiles Distance

Two particles are launched simultaneously from the exact same location with an identical initial speed  $v_0 = 25 \text{ m/s}$ . The first particle is projected vertically upward ( $\theta_1 = 90^\circ$ ), while the second particle is launched at an angle  $\theta_2 = 60^\circ$  above the horizontal.

Ignoring air resistance, calculate the distance between the two particles at time  $t = 1.70$  s after launch.



### PROBLEM 14

LEVEL II

#### Direct Aim at a Vertical Wall

A projectile is launched from ground level at an angle  $\theta = 30^\circ$  above the horizontal with an initial speed  $v_0 = 40$  m/s, aimed directly at a specific target located on a vertical wall at a horizontal distance  $d = 70$  m.

Neglecting air resistance, calculate the vertical deflection from the target (i.e., the vertical distance between the intended target point and the actual impact point on the wall).

### PROBLEM 15

LEVEL III

#### Cliff-Top Projectiles at Opposite Angles

Two identical balls are launched from the edge of a cliff at height  $h = 40$  m above level ground with the same initial speed  $v_0 = 30$  m/s. Ball 1 is launched at an angle  $\theta = 30^\circ$  above the horizontal, while Ball 2 is launched at  $\theta = 30^\circ$  below the horizontal.

- Find the difference in time of flight  $\Delta t = t_1 - t_2$  between the two balls hitting the ground.
- Find the horizontal distance between their impact points on the ground.
- Compare your answers for time of flight and horizontal distance with those of a projectile launched from level ground with speed  $v_0 = 30$  m/s at  $\theta = 30^\circ$  above the horizontal. What do you observe?

### PROBLEM 16

## LEVEL II

**Maximum Height to Range Ratio**

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A projectile is launched from level ground at an angle  $\theta$  above the horizontal. Let  $R$  be its horizontal range and  $H$  be its maximum height.

- (a) Derive an expression for the ratio  $H/R$  in terms of the launch angle  $\theta$ .
- (b) If a projectile is fired such that its horizontal range is equal to two times its maximum height ( $R = 2H$ ), find the launch angle  $\theta$ .

**PROBLEM 17**

## LEVEL II

**Fixed Initial Speed with Variable Launch Angle**

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A cannon fires projectiles at a fixed initial speed  $v_0$ , but the launch angle  $\theta$  can be varied. The maximum possible horizontal range achievable by the cannon is  $L$ .

Determine the horizontal range of a projectile fired at an angle  $\theta = 30^\circ$  above the horizontal in terms of  $L$ .

**PROBLEM 18**

## LEVEL II

**Level-Ground Projectile with Range 60m**

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A projectile is launched from level ground at an initial angle  $\theta = 45^\circ$  and achieves a horizontal range  $R = 60$  m. Neglecting air resistance:

- (a) Determine the initial speed  $v_0$  of the projectile.
- (b) Determine the total time of flight  $T$ .
- (c) How would the answers to (a) and (b) change if the range remained  $R = 60$  m, but the launch angle were increased to  $\theta > 45^\circ$ ? Explain physically.

**PROBLEM 19**

## LEVEL II

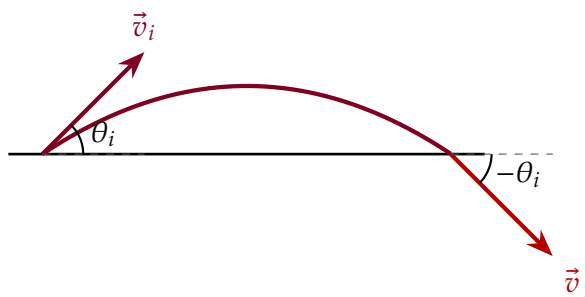
**Impact Velocity and Vector Direction**

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A particle is launched from level ground with an initial velocity  $\vec{v}_i$  at an angle  $\theta_i$  above the positive  $x$ -axis. When it returns to ground level, its final velocity vector is  $\vec{v}_f$ .

Expressed in terms of  $v_i$  and  $\theta_i$ :

- Find the angle between the initial velocity vector  $\vec{v}_i$  and the final velocity vector  $\vec{v}_f$ .
- Express the sum of velocity vectors  $\vec{v}_i + \vec{v}_f$  in unit vector notation ( $\hat{i}, \hat{j}$ ).



## PROBLEM 20

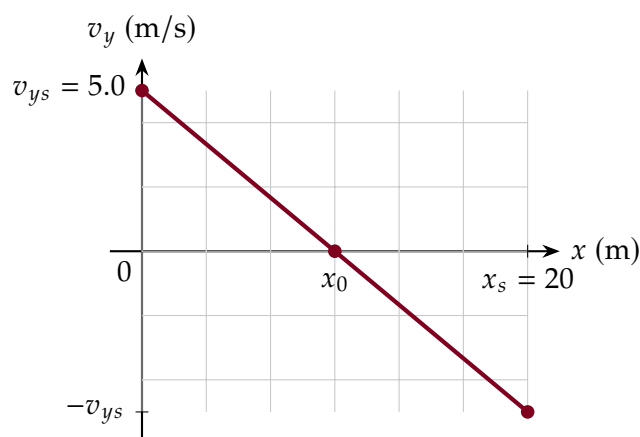
### LEVEL III

#### Vertical Component of Impact Velocity against a Wall

A ball is shot from level ground toward a vertical wall located at a distance  $x$  from the launch point. The figure below shows the vertical velocity component  $v_y$  of the ball just as it reaches the wall, plotted as a function of the distance  $x$ .

The graph scaling is defined by  $v_{ys} = 5.0$  m/s and  $x_s = 20$  m.

Determine the launch angle  $\theta_0$  of the ball.



# Hints

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The hints below correspond to the problems in the handout. They are provided separately so that students can work through the problem collection without immediately seeing the associated guidance.

@hintlist